Trade Validation and Exception Tracker

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What I built

A tracker that checks 2,000 simulated trades against counterparty and product reference tables before settlement, flags every exception, and shows how long each one has been open. The dataset was simulated using AI.

How it works

I built nine validation rules in Python (pandas) in Google Colab. They catch missing fields, a counterparty or product ID that is not in the reference table, a settlement date before the trade date, a settlement date on a weekend, a quantity above the product limit, a suspended counterparty, a trade currency that differs from the product currency, and duplicated trade IDs. Each exception is aged as of 30 September 2026 and assigned an escalation owner. I exported the results to Excel for the dashboard, and I rebuilt five of the rules in SQL (SQLite) to confirm that both methods return the same counts.

What it found

180 of the 2,000 trades had at least one problem. The exception list has 195 rows because 15 trade IDs appear twice, so both copies are listed. Duplicated IDs were the most common hit at 30 rows, followed by missing fields, unknown counterparties and oversized quantities at 25 each. Of the 195 exception rows, 111 (57%) had been open for more than ten days, which under my escalation rule would put them with the head of operations.

Trades keyed in by hand had an exception rate of 35.8% (126 of 352), against 4.2% (69 of 1,648) for electronic trades. I built that gap into the data, so it is not a finding about real banks. It shows what this tracker would surface if real data looked the same.

Assumptions

The age bands (0 to 2, 3 to 5, 6 to 10 and over 10 days) and the escalation owners are my own choices, not a real bank's policy. Settlement is treated as two business days after trade date, with weekends but no public holidays.

What I would change

If real data looked like this, I would stop the errors at the point of entry instead of finding them later. Manual booking screens should reject an unknown counterparty or product ID, block a settlement date earlier than the trade date, warn when a quantity is over the product limit, and refuse a trade ID that already exists. Every exception caught at entry is one that never reaches the morning queue. I would also ask who owns counterparty status updates, because a suspended firm should be blocked automatically and not found afterwards.